Synthetic Data Generation Framework

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Agenda

01. **Recap**

02. **Progress so far**

03. **System Overview**

04. **Key Technical Highlights**

05. **Evaluation Results**

06. **Next Steps**

Recap:

This iteration covers the completion of Stages 3, 4, and 5 planning.

- Built 5-stage sequential pipeline:
  - Project Metadata → Schedule → Task Summary → RFI → Budget
- Stage 1 complete: 1,000 synthetic projects using CTGAN and Gaussian Copula
- Stage 2 complete: Schedule baseline with SCD Type 2, rebaseline events via zero-inflated Poisson
- Stages 3, 4, 5 were planned, now implemented!

System Design

Synthetic Data Generation Framework

PROFILING LAYER

- **Distribution Capture**
  p5-p95, PMFs, Percentiles
- **Correlation Analysis**
  Joint pairs, Empirical PMFs
- **Business Rules**
  Domain Constraints

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GENERATION LAYER

Models

- **CTGAN**
  Deep neural modeling
- **Gaussian Copula**
  Stable with limited data
- **SDV Framework**
  Constraint enforcement

Statistical Approach

- **Hierarchical Fallback**
  SBU×Asset → SBU → Global
- **Soft Bounding**
  Jitter ±6-15% vs hard clamps
- **CDF Interpolation**
  Time-% spend curves

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VALIDATION LAYER

- **Correlation Checks**
  Field relationships preserved
- **Business Rules**
  Zero tolerance violations
- **FK Integrity**
  Cross-table validation
- **Train Synth / Test Real**
  ML model performance

Completed This Iteration

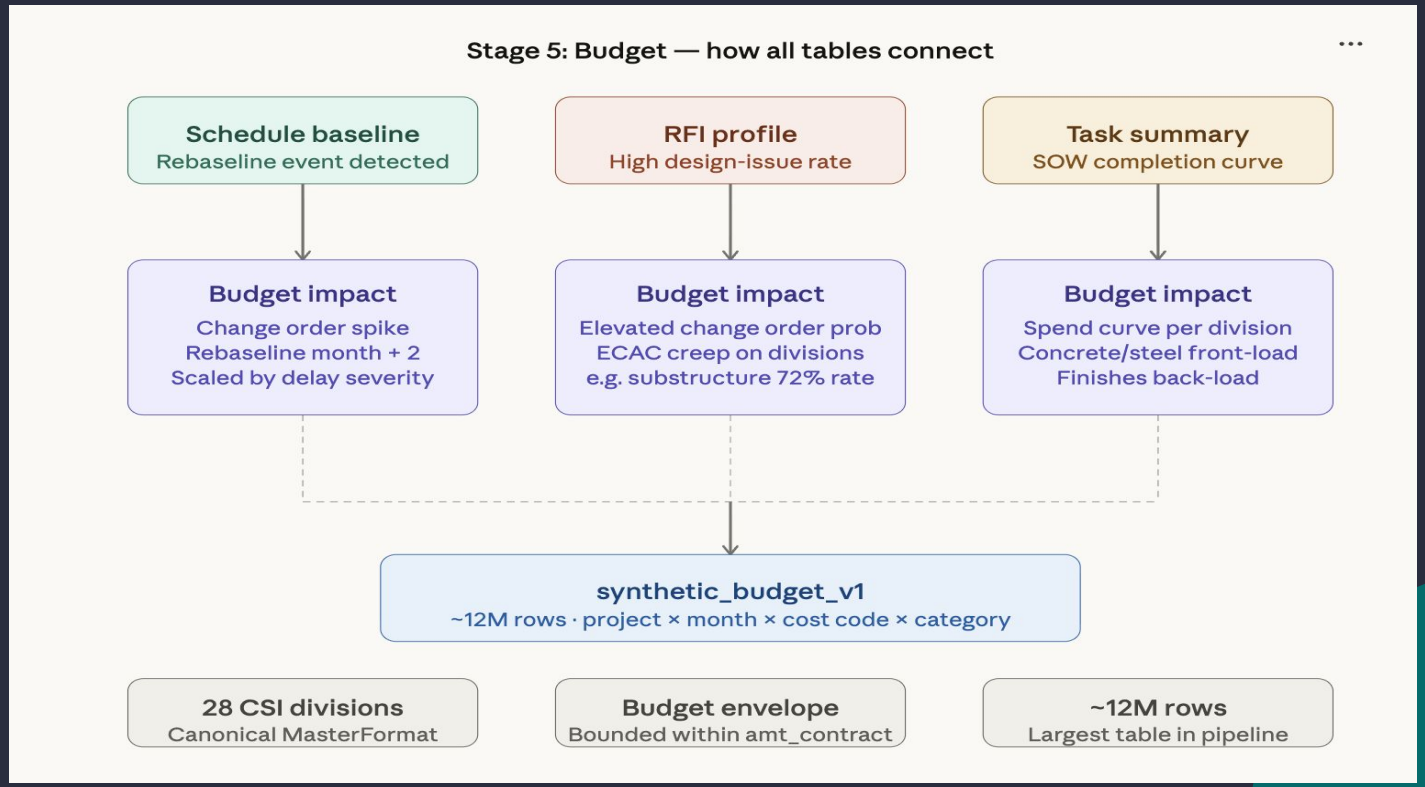
Stage 3 - Task Summary:

- 60,995 rows across 1,000 projects, zero date violations
- 84% SOW overlap modeled from real P6 data (construction is parallel, not sequential)
- Critical path percentages grounded in P6 zero-float analysis, 1 to 9% range per SOW
- Completion progression monotonically enforced across rebaseline versions

Stage 4 - RFI Records (Smoke Test Complete):

- 1,664 records for one project, including 141 chain follow-up RFIs
- Hybrid approach: structured fields from statistical profiles, text from LLM (Anthropic API)
- RFI volume multiplier of 1.08x applied within 30 days of a rebaseline event

Key Technical Highlight



Evaluation and Validation

Statistical fidelity:

- KS test: statistic below 0.1 threshold for all continuous fields
- Chi-squared: categorical distributions preserved across SBU, asset class, SOW type
- Spearman correlation: contract value, gross area, duration relationships maintained

Structural integrity:

- Zero date violations across 60,995 task summary rows
- FK integrity verified across all five tables
- Budget envelope: total original budget per project within 105% of contract value
- Monotonicity enforced on cost and completion time series

Next steps

01.

Run budget profiling script and smoke test on SYN-COM-000031

02.

Full budget generation across 1,000 projects after supervisor sign-off

03.

Full RFI generation at scale pending cost approval; tighten RFI to budget causal signal using actual records

04.

Lifecycle evaluation framework, cross-table interaction analysis, white paper, and final report submission

Any questions? Ask away!

Thank you